

Notes 4 - Databases and SQL

Math 3190 - Fundamentals of Data Science

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Section 1

Databases and SQL

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Databases and SQL

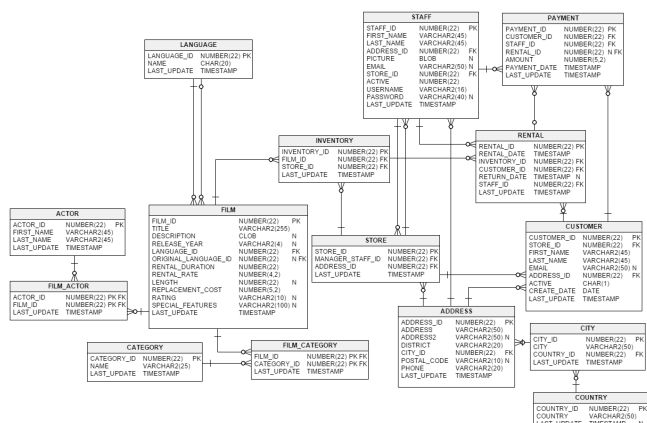
Intro to Database Design

In many businesses and government agencies, data files are not directly accessible. Instead, data are stored in databases that are usually in the cloud. The most common type of database is a *relational database*. These types of databases are made up of **tables**, which are very similar to data frames in **R**. These tables usually have at least one variable in common with at least one other table that can be used to connect them.

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Databases and SQL

Relational Database Example



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Databases and SQL

SQL Intro

The most common way to access information stored in relational databases is with SQL.

SQL stands for structured query language and is ubiquitous in business, government, and academia where large data sets are used. SQL is usually either pronounced like "sequel" or the letters are individually said "S-Q-L".

We are going to go over the common SQL commands and then look at how to access SQL databases and use SQL within **R**.

SQL is one of the most basic "coding languages" there is because of how similar the statements are to spoken English.

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SELECT, FROM, and WHERE Statements

The backbone of SQL are the SELECT, FROM, and WHERE statements.

- SELECT is very much like the `select()` function in the tidyverse (dplyr package). It selects the columns or variables from the database. You can use "*" to select all columns.
- FROM is the statement that specifies the table in the database from which we are taking the data.
- WHERE is similar to the `filter()` function in the tidyverse. This will subset the table based on some criterion or criteria.

Note: these words do not *need* to be written in all capital letters, but it is the convention to type them in all caps for readability.

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SQL Syntax

The basic syntax for SQL statements is as follows:

```
SELECT column1, column2, ...
FROM table_name
WHERE condition;
```

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SQL Example

Suppose we have a database named sakila that models a DVD rental store, featuring things like films, actors, film-actor relationships. In this database are many tables, one of which is "film". Here is some information about the film table:

Let's run some SQL queries with this table.

film	
film_id	smallint
title	varchar
description	text
release_year	year
language_id	tinyint
original_language_id	tinyint
rental_duration	tinyint
rental_rate	decimal
length	smallint
replacement_cost	decimal
rating	enum
special_features	set
last_update	timestamp

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SQL Query 1

```
SELECT title, release_year, rating
FROM film
WHERE rating = 'PG';
```

Table 1: My Caption

title	release_year	rating
ACADEMY DINOSAUR	2006	PG
AGENT TRUMAN	2006	PG
ALASKA PHANTOM	2006	PG
ALI FOREVER	2006	PG
AMADEUS HOLY	2006	PG

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WHERE Statement Commands

The following can be used in a WHERE statement:

Operator	Description
=	Equal
>	Greater than
<	Less than
>=	Greater than or equal
<=	Less than or equal
<> or !=	Not equal
AND	Restricts the output based on another criterion.
OR	Expands the output based on another criterion.
NOT	Gives the opposite of the criterion.
BETWEEN	Between a certain range.
IN	To specify multiple possible values for a column.
LIKE	Search for a pattern.

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SQL Query 2: BETWEEN

The syntax for the BETWEEN statement is

```
WHERE column_name BETWEEN value1 AND value2
SELECT film_id, title, length, rental_duration
FROM film
WHERE film_id BETWEEN 13 AND 14
OR film_id BETWEEN 21 AND 23;
```

Table 2: Displaying records 1 - 5

film_id	title	length	rental_duration
13	ALI FOREVER	150	4
14	ALICE FANTASIA	94	6
21	AMERICAN CIRCUS	129	3
22	AMISTAD MIDSUMMER	85	6
23	ANACONDA CONFESSIONS	92	3

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SQL Query 3: IN

The IN operator is a shorthand for multiple OR conditions.

```
SELECT film_id, title, length, rental_duration
FROM film
WHERE film_id IN ("1", "50", "100", "613");
```

Table 3: Displaying records 1 - 4

film_id	title	length	rental_duration
1	ACADEMY DINOSAUR	86	6
50	BAKED CLEOPATRA	182	3
100	BROOKLYN DESERT	161	7
613	MYSTIC TRUMAN	92	5

Note: the quotes aren't needed for numbers, but can still be used.

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SQL Query 4: LIKE

There are two wildcards often used in conjunction with the LIKE operator:

- The percent sign % represents zero, one, or multiple characters.
- The underscore sign _ represents one single character. This can be repeated to stand for multiple characters.

```
SELECT film_id, title, length, rental_duration
FROM film
WHERE title LIKE "b%" AND length >= 175;
```

Table 4: Displaying records 1 - 4

film_id	title	length	rental_duration
50	BAKED CLEOPATRA	182	3
61	BEAUTY GREASE	175	5
88	BORN SPINAL	179	7
94	BRAVEHEART HUMAN	176	7

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SQL Query 5: LIKE

```
SELECT film_id, title, length, rental_duration
FROM film
WHERE title LIKE "__ab%";
```

Table 5: 3 records

film_id	title	length	rental_duration
9	ALABAMA DEVIL	114	3
34	ARABIA DOGMA	62	6
773	SEABISCUIT PUNK	112	6

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More on SELECT Statments

In the SELECT statement, instead of simply putting column names, we can also use the following functions:

- MIN() returns the minimum value of a numeric column.
- MAX() returns the maximum value of a numeric column.
- COUNT() returns the number of rows that matches a specified criterion.
- SUM() returns the total sum of a numeric column.
- AVG() returns the average value of a numeric column.

Additionally, we can use aliases to rename columns with the AS statement.

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SQL Query 6: SELECT Functions

```
SELECT AVG(length) AS "Average Length"
FROM film;
```

Table 6: 1 records

Average Length
115.272

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SQL Query 7: SELECT Functions

```
SELECT COUNT(title) AS "Number of A or B Movies"
FROM film
WHERE title LIKE "a%" OR title LIKE "b%";
```

Table 7: 1 records

Number of A or B Movies
109

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SQL Query 8: SELECT Functions

The following will return all the films with a length higher than the average length.

```
SELECT title, length
FROM film
WHERE length > (SELECT AVG(length) FROM film);
```

Table 8: Displaying records 1 - 5

title	length
AFFAIR PREJUDICE	117
AFRICAN EGG	130
AGENT TRUMAN	169
ALAMO VIDEOTAPE	126
ALASKA PHANTOM	136

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ORDER BY Statement

We can also use the ORDER BY statement to change the order in which the data table is given.

The syntax for ORDER BY is

```
ORDER BY column1, column2, ... columnN
```

We can put a ASC for ascending or DESC for descending values after any one of the columns we are ordering by.

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SQL Query 9: ORDER BY Statement

```
SELECT title, length
FROM film
WHERE length > (SELECT AVG(length) FROM film)
ORDER BY length ASC, title DESC;
```

Table 9: Displaying records 1 - 5

title	length
WORDS HUNTER	116
MADIGAN DORADO	116
INSTINCT AIRPORT	116
DUCK RACER	116
RESURRECTION SILVERADO	117

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GROUP BY Statement

The GROUP BY statement groups rows that have the same values into summary rows. It is often used with aggregate functions (COUNT(), MAX(), MIN(), SUM(), AVG()) to group the result-set by one or more columns.

Syntax:

```
SELECT column_name(s)
FROM table_name
WHERE condition
GROUP BY column_name(s)
ORDER BY column_name(s);
```

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SQL Query 10: GROUP BY Statement

```
SELECT AVG(length), rating
FROM film
WHERE release_year = 2006
GROUP BY rating
ORDER BY rating;
```

Table 10: Displaying records 1 - 5

AVG(length)	rating
111.0506	G
113.2286	NC-17
112.0052	PG
120.4439	PG-13
118.6615	R

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SELECT DISTINCT Statement

The SELECT DISTINCT statement is used to return only distinct (different) values.

Syntax:

```
SELECT DISTINCT column1, column2, ...
FROM table_name;
```

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SQL Query 11: DISTINCT Statement

```
SELECT DISTINCT release_year
FROM film;
```

Table 11: 1 records

release_year
2006

So this database only has films from the year 2006.

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UNION Statement

The UNION operator is used to combine the result-set of two or more SELECT statements.

Syntax:

```
SELECT column_name(s) FROM table1
UNION
SELECT column_name(s) FROM table2;
```

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SQL Query 12: UNION Statement

```
SELECT AVG(length) AS "Average Length and Payment Amount"
FROM film
UNION
SELECT AVG(amount)
FROM payment;
```

Table 12: 2 records

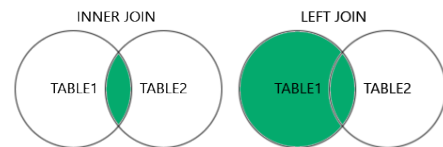
Average Length and Payment Amount
4.200667
115.272000

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JOIN Statements

A JOIN clause is used to combine rows from two or more tables, based on a related column between them. There are four types of JOIN statements:

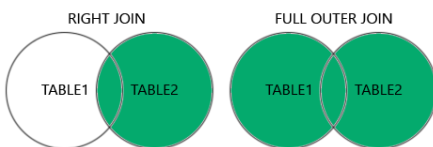
- (INNER) JOIN: Returns records that have matching values in both tables.
- LEFT (OUTER) JOIN: Returns all records from the left table, and the matched records from the right table.



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JOIN Statements

- RIGHT (OUTER) JOIN: Returns all records from the right table, and the matched records from the left table.
- FULL (OUTER) JOIN: Returns all records when there is a match in either left or right table.



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JOIN Statements

The syntax for JOIN statements is as follows:

```
SELECT column_name(s)
FROM table1
(JOIN STATEMENT) table2
ON table1.column_name = table2.column_name;
```

Replace the (JOIN STATEMENT) with INNER JOIN, LEFT JOIN, RIGHT JOIN, or FULL JOIN.

You can select columns from either table, so it is common to denote which table we are selecting from by using table.column_name.

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JOIN Examples

Consider the following two tables called Student and StudentCourse, respectively.

ROLL_NO	NAME	ADDRESS	PHONE	Age
1	HARSH	DELHI	XXXXXXXXXX	18
2	PRATIK	BIHAR	XXXXXXXXXX	19
3	RIYANKA	SILIGURI	XXXXXXXXXX	20
4	DEEP	RAMNAGAR	XXXXXXXXXX	18
5	SAPTARHI	KOLKATA	XXXXXXXXXX	19
6	DHANRAJ	BARABAJAR	XXXXXXXXXX	20
7	ROHIT	BALURGHAT	XXXXXXXXXX	18
8	NIRAJ	ALIPUR	XXXXXXXXXX	19

COURSE_ID	ROLL_NO
1	1
2	2
2	3
3	4
1	5
4	9
5	10
4	11

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JOIN Example: INNER JOIN

```
SELECT StudentCourse.COURSE_ID, Student.NAME, Student.AGE
FROM Student
INNER JOIN StudentCourse
ON Student.ROLL_NO = StudentCourse.ROLL_NO;
```

COURSE_ID	NAME	Age
1	HARSH	18
2	PRATIK	19
2	RIYANKA	20
3	DEEP	18
1	SAPTARHI	19

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JOIN Example: LEFT JOIN

```
SELECT Student.NAME, StudentCourse.COURSE_ID
FROM Student
LEFT JOIN StudentCourse
ON StudentCourse.ROLL_NO = Student.ROLL_NO;
```

NAME	COURSE_ID
HARSH	1
PRATIK	2
RIYANKA	2
DEEP	3
SAPTARHI	1
DHANRAJ	NULL
ROHIT	NULL
NIRAJ	NULL

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JOIN Example: RIGHT JOIN

```
SELECT Student.NAME, StudentCourse.COURSE_ID
FROM Student
RIGHT JOIN StudentCourse
ON StudentCourse.ROLL_NO = Student.ROLL_NO;
```

NAME	COURSE_ID
HARSH	1
PRATIK	2
RIYANKA	2
DEEP	3
SAPTARHI	1
NULL	4
NULL	5
NULL	4

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JOIN Example: FULL JOIN

```
SELECT Student.NAME, StudentCourse.COURSE_ID
FROM Student
FULL JOIN StudentCourse
ON StudentCourse.ROLL_NO = Student.ROLL_NO;
```

NAME	COURSE_ID
HARSH	1
PRATIK	2
RIYANKA	2
DEEP	3
SAPTARHI	1
DHANRAJ	NULL
ROHIT	NULL
NIRAJ	NULL
NULL	4
NULL	5
NULL	4

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Section 2

SQL in Quarto

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SQL in Quarto

In Quarto, you can insert an SQL code chunk.

```
```{sql}
--| connection: connection_name
```
```

Be aware that the setup comments for SQL code chunks are `--|` instead of `#|` because an SQL comment is `--` whereas **R** and Python comments are both `#`.

We will need to establish a connection before using the SQL code chunk, though. This connection must be to a database, either remote or local.

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Define an SQL Connection in R

To connect to a remote database, we can use the following code chunk in Quarto. We will need a few SQL and database packages, though. Some common ones are DBI, RMySQL, and RSQL, and RSQLite. This database is a MySQL database, so we will need the RMySQL library.

```
```{r}
#| echo: true
#| eval: true
library(DBI)
library(RMySQL)
con <- dbConnect(RMySQL::MySQL(),
 dbname = "sakila",
 host = "relational.fit.cvut.cz",
 port = 3306,
 user = "guest",
 password = "relational")
```
```

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Define an SQL Connection in R

Note: this website for the sakilla database has since gone down since these notes were created. A remote database I found that is still up (as of January 31, 2024) can be accessed using the code on the following slide:

```
```{r}
#| echo: true
#| eval: true
con = dbConnect(RMySQL::MySQL(),
 dbname='Rfam',
 host='mysql-rfam-public.ebi.ac.uk',
 port=4497,
 user='rfamro',
 password='')
```
```

The following slides will still use the sakila database as an example.

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Define an SQL Connection in R

Then we can use that connection in our SQL code chunk.

```
```{sql}
--| connection: con
--| echo: true
--| eval: true
--| max-print: 3
SELECT title, release_year, rating
FROM film
WHERE rating = 'PG';
```
```

Table 13: Displaying records 1 - 3

| title | release_year | rating |
|------------------|--------------|--------|
| ACADEMY DINOSAUR | 2006 | PG |
| AGENT TRUMAN | 2006 | PG |
| ALASKA PHANTOM | 2006 | PG |

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Creating a Data Frame in R from SQL

Instead of printing the table, we can store the table we created from our SQL command as a data frame using the `output.var` argument in the SQL code chunk:

```
```{sql}
--| connection: con
--| output.var: pg_movies
SELECT title, release_year, rating
FROM film
WHERE rating = 'PG';
```
```

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Using SQL Table in R

Now we can access the `pg_movies` variable in **R**.

```
pg_movies <- tibble(pg_movies)
head(pg_movies, 5)
# A tibble: 5 x 3
  title                release_year rating
<chr>                <dbl> <chr>
1 ACADEMY DINOSAUR      2006 PG
2 AGENT TRUMAN          2006 PG
3 ALASKA PHANTOM        2006 PG
4 ALI FOREVER           2006 PG
5 AMADEUS HOLY          2006 PG

dim(pg_movies)
[1] 194 3
```

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Using Tidyverse Functions

The same thing can be accomplished if we had read in the film table as a data frame or tibble.

```
movies <- read.csv("film.csv")
pg_movies <- movies |>
  filter(rating == "PG") |>
  select(title, release_year, rating) |> tibble()
head(pg_movies, 5)

# A tibble: 5 x 3
  title                release_year rating
  <chr>                <int> <chr>
1 ACADEMY DINOSAUR      2006 PG
2 AGENT TRUMAN          2006 PG
3 ALASKA PHANTOM         2006 PG
4 ALI FOREVER            2006 PG
5 AMADEUS HOLY          2006 PG
```

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Using Tidyverse Functions

The following statements and functions are equivalent:

| SQL | R Tidyverse |
|-------------------|--------------|
| SELECT | select() |
| WHERE | filter() |
| ORDER BY | arrange() |
| GROUP BY | group_by() |
| INNER JOIN | inner_join() |
| LEFT JOIN | left_join() |
| RIGHT JOIN | right_join() |
| FULL (OUTER) JOIN | full_join() |

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Using R Objects in SQL Chunks

If you need to use the values of an **R** variables into SQL queries, you can do so by prefacing **R** variable references with a `?`. For example:

```
```{r}
movie_length <- 130
```

```{sql}
--| connection: con
--| max-print: 5
SELECT title, length
FROM film
WHERE length >= ?movie_length
```
```

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Using R Objects in SQL Chunks

Table 14: Displaying records 1 - 5

| title | length |
|-----------------|--------|
| AFRICAN EGG | 130 |
| AGENT TRUMAN | 169 |
| ALASKA PHANTOM | 136 |
| ALI FOREVER | 150 |
| ALLEY EVOLUTION | 180 |

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Close Session

When we are finished with connecting to the database, it is best to close the connection using the following line:

```
dbDisconnect(con) # Disconnects from server.
```

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Other SQL Uses

There are lots of other things we can do using SQL. We can create tables, create databases, drop tables, insert rows and columns into tables, and more.

For this class, however, we will leave it there since most of the time, we will simply be accessing databases for data analysis rather than editing them.

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Section 3

Local Databases

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Simulating SQL Database with Local Files in Memory

To simulate a connection to a database, we can use the following:

```
con <- dbConnect(RSQLite::SQLite(), ":memory:")
sakila_files <- list.files("../Data/Sakila DB Data")
for (file in sakila_files) {
  data_file <- read.csv(paste0("../Data/Sakila DB Data/", file))
  # Add data frame as a table
  dbWriteTable(con, substr(file, 1, nchar(file) - 4), data_file)
  rm(data_file)
}
dbListTables(con)
```

| | | | |
|---------------------|-------------|-------------|--------------|
| [1] "actor" | "address" | "category" | "city" |
| [5] "country" | "customer" | "film" | "film_actor" |
| [9] "film_category" | "film_text" | "inventory" | "language" |
| [13] "payment" | "rental" | "staff" | "store" |

These tables are not stored in R's memory, though!

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Simulating SQL Database with Local Files

Table 15: Displaying records 1 - 5

| title | length |
|------------------|--------|
| AFFAIR PREJUDICE | 117 |
| AFRICAN EGG | 130 |
| AGENT TRUMAN | 169 |
| ALAMO VIDEOTAPE | 126 |
| ALASKA PHANTOM | 136 |

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Connect to Local Data Databases

If we have an SQL-type file (like one that ends in .sqlite), then we can connect to it without reading it into R's memory.

For example, I have a database called "portal_mammals.sqlite" (downloadable from [here](#)). If it is in my working directory, I can connect to this database using

```
mammals_con <- DBI::dbConnect(RSQLite::SQLite(),
                              "portal_mammals.sqlite")
DBI::dbListTables(mammals_con) # List all database tables
```

| | | |
|-------------|-----------|-----------|
| [1] "plots" | "species" | "surveys" |
|-------------|-----------|-----------|

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Connect to Local Data Databases

Then I can run SQL queries using the dbGetQuery() function:

```
dbGetQuery(mammals_con, "SELECT * FROM surveys;") |> head()
```

| | record_id | month | day | year | plot_id | species_id | sex | hindfoot_length |
|---|-----------|-------|-----|------|---------|------------|-----|-----------------|
| 1 | 1 | 7 | 16 | 1977 | 2 | NL | M | |
| 2 | 2 | 7 | 16 | 1977 | 3 | NL | M | |
| 3 | 3 | 7 | 16 | 1977 | 2 | DM | F | |
| 4 | 4 | 7 | 16 | 1977 | 7 | DM | M | |
| 5 | 5 | 7 | 16 | 1977 | 3 | DM | M | |
| 6 | 6 | 7 | 16 | 1977 | 1 | PF | M | |

or I can connect to the database in Quarto using

```
```{sql}
--| connection: mammals_con
--| max-print: 5
SELECT * FROM surveys;
```
```

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Connect to Local Data Databases

Select from December 2000:

```
SELECT record_id, month, day, year, species_id, sex, weight
FROM surveys
WHERE month = 12 AND year = 2000;
```

Table 16: Displaying records 1 - 5

| record_id | month | day | year | species_id | sex | weight |
|-----------|-------|-----|------|------------|-----|--------|
| 31616 | 12 | 22 | 2000 | DO | M | 52 |
| 31617 | 12 | 22 | 2000 | PB | F | 26 |
| 31618 | 12 | 22 | 2000 | OT | M | 24 |
| 31619 | 12 | 22 | 2000 | PB | F | 25 |
| 31620 | 12 | 22 | 2000 | DM | M | 38 |

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Create a Local Database

We can create a .sqlite file from a .csv using the `csv_to_sqlite()` function from the `inborutils` package. This is not available on the CRAN, but can be installed using

`devtools::install_github("inbo/inborutils")`.

This function reads a csv in smaller chunks (50,000 lines at a time by default) and writes it to a .sqlite file so we can connect to it as a database.

Example:

```
library(inborutils)
csv_to_sqlite(csv_file = "combined.csv",
              sqlite_file = "combined.sqlite",
              table_name = "combined")

combined_con <- DBI::dbConnect(RSQLite::SQLite(),
                              "combined.sqlite")
dbGetQuery(combined_con, "SELECT * FROM combined") |> head()
```

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Create a Local Database

| record_id | month | day | year | plot_id | species_id | sex |
|-----------|-------|-----|------|---------|------------|---------|
| 1 | 1 | 7 | 16 | 1977 | 2 | NL M |
| 2 | 72 | 8 | 19 | 1977 | 2 | NL M |
| 3 | 224 | 9 | 13 | 1977 | 2 | NL <NA> |
| 4 | 266 | 10 | 16 | 1977 | 2 | NL <NA> |
| 5 | 349 | 11 | 12 | 1977 | 2 | NL <NA> |
| 6 | 363 | 11 | 12 | 1977 | 2 | NL <NA> |

| hindfoot_length | weight | genus | species | taxa | plot_type |
|-----------------|--------|-------|------------------|--------|-----------|
| 1 | 32 | NA | Neotoma albigula | Rodent | Control |
| 2 | 31 | NA | Neotoma albigula | Rodent | Control |
| 3 | NA | NA | Neotoma albigula | Rodent | Control |
| 4 | NA | NA | Neotoma albigula | Rodent | Control |
| 5 | NA | NA | Neotoma albigula | Rodent | Control |
| 6 | NA | NA | Neotoma albigula | Rodent | Control |

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Create a Local Database

You can add another table to the same .sqlite file using `csv_to_sqlite()` as long as a different table name is specified.

```
csv_to_sqlite(csv_file = "actor.csv",
              sqlite_file = "sakila.sqlite",
              table_name = "actor")
csv_to_sqlite(csv_file = "film.csv",
              sqlite_file = "sakila.sqlite",
              table_name = "film")
```

The above code will create the `sakila.sqlite` file in the first function call with the table "actor". The second function call will then add the "film" table to that same file.

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Convert from dplyr Functions to SQL

The `dbplyr` package is an add-on to the `dplyr` package that allows you to use remote database tables as if they are in-memory data frames by automatically converting dplyr code into SQL. We can install it in the normal way: `install.packages("dbplyr")`. If you don't install it, you will be prompted to when you do something that requires it.

Example:

```
# Access the SQL table "surveys" from the mammals_con connection
mammal_surveys <- tbl(mammals_con, "surveys")
mammal_surveys
```

```
# Source:   table<'surveys'> [?? x 9]
# Database: sqlite 3.51.0 [/Users/rickbrown/My Drive/Math 3190 - Fun
  record_id month   day   year plot_id species_id sex
    <int> <int> <int> <int>   <int>   <chr>   <chr>
1         1     7    16  1977      2    NL      M
2         2     7    16  1977      3    NL      M
3         3     7    16  1977      2    DM      F
4         4     7    16  1977      7    DM      M
```

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Convert from dplyr Functions to SQL

```
mammal_summary <- mammal_surveys |>
  filter(month == 7 & year == 1992) |>
  group_by(species_id) |>
  summarize(avg_weight = mean(weight, na.rm = TRUE)) |>
  arrange(desc(avg_weight))
head(mammal_summary)

# Source:   SQL [?? x 2]
# Database: sqlite 3.51.0 [/Users/rickbrown/My Drive/Math 3190 - Fun
# Ordered by: desc(avg_weight)
  species_id avg_weight
    <chr>      <dbl>
1 NL         131.
2 DS         120
3 SO         57.5
4 DO         44.6
5 DM         43.4
6 NL         32.4
```

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Convert from dplyr Functions to SQL

The `show_query()` function will show an SQL query that will accomplish the same thing the dplyr functions did to create the object. Sometimes these queries look a bit strange, but they work!

```
show_query(mammal_summary)

<SQL>
SELECT `species_id`, AVG(`weight`) AS `avg_weight`
FROM (
  SELECT `surveys`.*
  FROM `surveys`
  WHERE (`month` = 7.0 AND `year` = 1992.0)
) AS `q01`
GROUP BY `species_id`
ORDER BY `avg_weight` DESC
```

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Same thing with SQL

```
SELECT `species_id`, AVG(`weight`) AS `avg_weight`
FROM (
  SELECT `surveys`.*
  FROM `surveys`
  WHERE (`month` = 7.0 AND `year` = 1992.0)
) AS `q01`
GROUP BY `species_id`
ORDER BY `avg_weight` DESC
```

Table 17: Displaying records 1 - 10

species_id	avg_weight
NL	130.66667
DS	120.00000
SO	57.50000
DO	44.60000
DM	43.42500

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Object Sizes

What is especially amazing about this is even though we are working with an object called `mammal_surveys`, it is actually not read into memory. We can see that here using the `obj_size()` function in the `lobstr` package:

```
lobstr::obj_size(mammal_surveys)
```

5.98 kB

```
# collect function brings the object into memory
mammal_surveys2 <- collect(mammal_surveys)
lobstr::obj_size(mammal_surveys2)
```

1.57 MB

So, being an **R** wizard also makes you an SQL wizard (by proxy)!

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Close Connection

As always, we should not forget to disconnect when we are done.

```
dbDisconnect(con)
dbDisconnect(combined_con)
dbDisconnect(mammals_con)
```

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Conclusion

- Knowing SQL is a great skill to not only access databases remotely, but also to be able to work with large files locally without reading them into **R**.
- Even though tools exist to convert **R** code to SQL queries, it is still important to know both to be able to handle working with any database or data set.

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Session info

```
R version 4.5.2 (2025-10-31)
Platform: aarch64-apple-darwin20
Running under: macOS Sequoia 15.7.2

Matrix products: default
BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/libBLAS.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK version 3.12.0

locale:
 [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/Denver
tzcode source: internal

attached base packages:
[1] stats      graphics  grDevices  utils      datasets
[6] methods   base

other attached packages:
[1] inborutils_0.4.0 lubridate_1.9.4 forcats_1.0.1
[4] stringr_1.6.0    dplyr_1.1.4    purrr_1.2.0
[7] readr_2.1.6      tidyr_1.3.2    tibble_3.3.0
[10] ggplot2_4.0.1    tidyverse_2.0.0 RMySQL_0.11.1
[13] DBI_1.2.3

loaded via a namespace (and not attached):
[1] gtable_0.3.6      xfun_0.55
[3] htmlwidgets_1.6.4 tzdb_0.5.0
[5] vctrs_0.6.5       tools_4.5.2
[7] crosstalk_1.2.2   generics_0.1.4
[9] parallel_4.5.2    curl_7.0.0
[11] proxy_0.4-29      RSQLite_2.4.4
[13] blob_1.2.4        pkgconfig_2.0.3
```

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